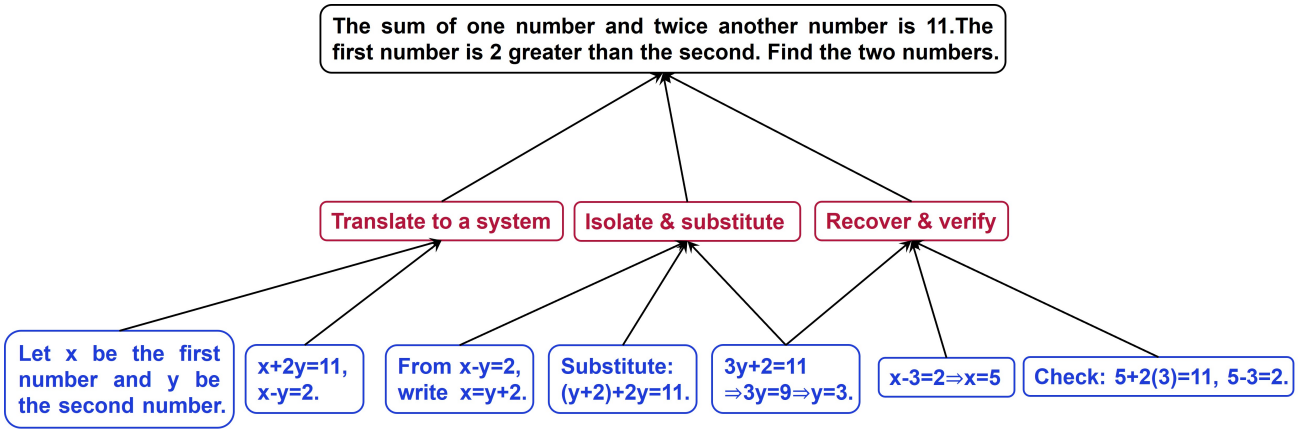

Figure H6. **A concrete algebra example illustrating skills, routing mechanisms, and identification conditions (Figure 2, Condition C.1).** The **black box** is the problem descriptor $\mathbf{P}$; the **red boxes** are latent selection variables; the **blue boxes** are observed trace tokens. **Neighborhood coverage** (Condition C.1-v): If "Translate to a system" only ever preceded "Isolate & substitute," it could be mistaken for one fused routine; seeing it also before "Eliminate by subtraction" ($(x+2y)-(x-y)=11-2 \Rightarrow 3y=9 \Rightarrow y=3$) provides a distinct neighborhood that isolates it as reusable. **Local distinguishability** (Condition C.1-iii,iv): "Substitute: $(y+2)+2y=11$" depends only on "Isolate & substitute" (no other red box), serving as its *anchor*; choosing elimination instead would produce a visibly different token, so distinct module states remain separable. **Restricted choice sets** (Condition C.1-ii): After "Substitute: $(y+2)+2y=11$," the algebraically valid continuation is "$3y+2=11 \Rightarrow 3y=9 \Rightarrow y=3$"; other token sequences would violate the problem's constraints, so valid traces form a strict subset of all combinatorial possibilities.

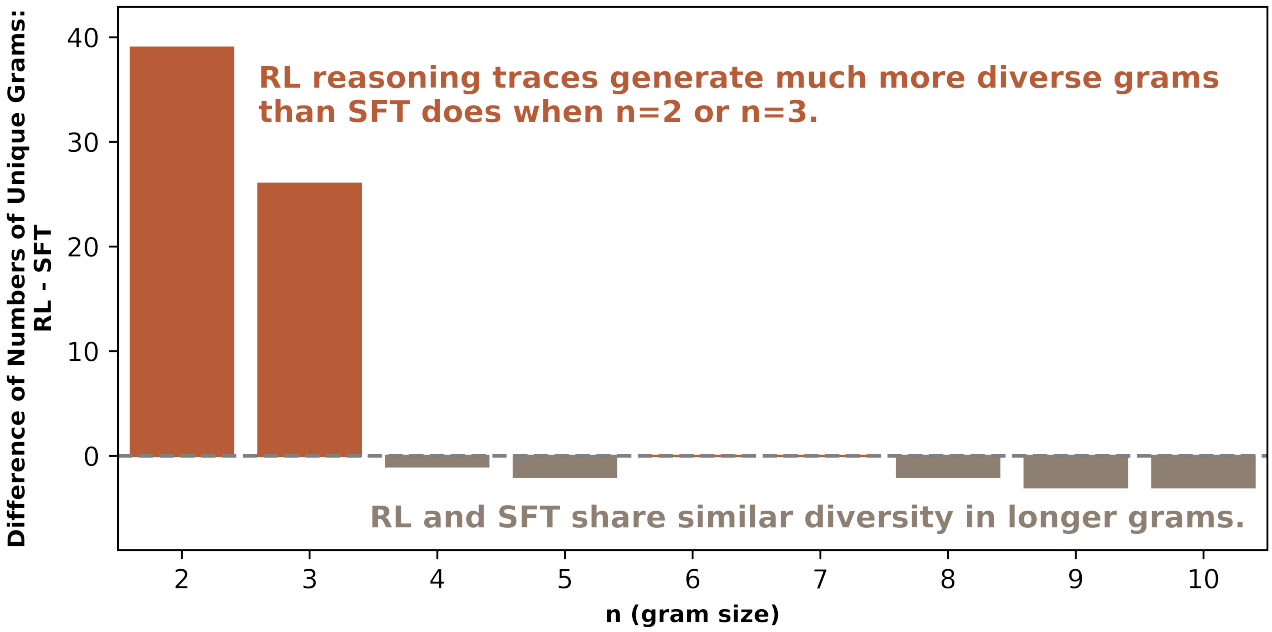

Figure H7. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n=2$ and $n=3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.